Prompt 3 - trial 1 (seed 0) - minimal

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```


![image-123.png](./image-123.png)


```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

Train a GNN on a compound activity dataset. Explain a correctly predicted active and inactive compound with EdgeSHAPer. Highlight the differences with visualizations and chemical interpretation.

Assistant

I'll help you train a GNN model on a compound activity dataset and perform EdgeSHAPer explainability analysis on both active and inactive compounds! Let's start by exploring the available datasets. 🔍

(Calling tool `find_datasets`)

Assistant

Great! I found several datasets. Let me load the binary activity dataset for our GNN training. 📊

(Calling tool `load_dataset`)

Assistant

Let me use the correct label column. I'll use `class_label` as the target.

(Calling tool `load_dataset`)

Assistant

Excellent! Dataset loaded with 8,348 compounds. Now let's split it for GNN training. 🔄

(Calling tool `split_dataset`)

Assistant

Dataset split successfully! Now let me prepare the GNN dataset from the SMILES. 🧪

(Calling tool `prepare_gnn_dataset`)

Assistant

GNN dataset prepared successfully! Now let's train the GCN model. 🚀

(Calling tool `train_gnn_model_mcp`)

Assistant

GNN training submitted! Let me check the training status. ⏳

(Calling tool `check_gnn_training`)

Assistant

Model is still training. Let me check again in a moment. ⏳

(Calling tool `check_gnn_training`)

Assistant

Still training... ⏳ Let me check again.

(Calling tool `check_gnn_training`)

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(Calling tool `check_gnn_training`)

Assistant

Excellent! 🎉 The GNN model has been trained successfully with impressive results:

Metric	Value
Validation Accuracy	95.33%
Test Accuracy	95.21%
Training Samples	6,678
Validation Samples	835
Test Samples	835

Now let's select correctly predicted compounds for EdgeSHAPer analysis! Let me start with an **active compound** (class 1). 🧪

(Calling tool `select_compound_for_edgeshaper`)

Assistant

Great! Found an **active compound** with 98.6% confidence. Now let me select an **inactive compound** (class 0). ⚗️

(Calling tool `select_compound_for_edgeshaper`)

Assistant

Perfect! Now I have both compounds:

Property	Active Compound	Inactive Compound
SMILES	`C#Cc1ccc2sc(C(=O)NCCCCN3CCN(c4ccccc4OC)CC3)cc2c1`	`COc1cccc(-c2ccc3c(NC(=O)c4ccc(C(=O)NO)cc4)n[nH]c3c2)c1F`
True Label	1 (Active)	0 (Inactive)
Predicted Label	1 (Active)	0 (Inactive)
Confidence	98.6%	99.6%

Now let me run EdgeSHAPer on the **active compound** first. 🔍

(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

EdgeSHAPer completed for the active compound! Now let me visualize the results and analyze the **inactive compound**. 📊

(Calling tool `visualize_edgeshaper_results`)

Assistant

Now let me display the EdgeSHAPer visualization for the **active compound** and analyze the **inactive compound**. 🖼️

(Calling tool `show_plot`)

Assistant

Now let me analyze the **inactive compound** with EdgeSHAPer. 🔬

(Calling tool `explain_gnn_with_edgeshaper`)

Assistant

Excellent! EdgeSHAPer analysis for the inactive compound is complete. Let me visualize it. 📊

(Calling tool `visualize_edgeshaper_results`)

Assistant

(Calling tool `show_plot`)

Assistant

Now let me get the detailed bond environment analysis for both compounds to provide comprehensive chemical interpretation. 🔬

(Calling tool `get_edgeshaper_bond_environments`)

Assistant

Now let me get the bond environments for the **inactive compound**. 🔬

(Calling tool `get_edgeshaper_bond_environments`)

Assistant

Now I have comprehensive EdgeSHAPer analysis for both compounds. Let me provide a detailed summary with chemical interpretation. 📊

🧪 GNN Model Training Summary

Metric	Value
Model Architecture	GCN (Graph Convolutional Network)
Training Samples	6,678
Validation Samples	835
Test Samples	835
Validation Accuracy	**95.33%**
Test Accuracy	**95.21%**

🔍 EdgeSHAPer Analysis: Active vs Inactive Compounds

1. ACTIVE Compound (Class 1)

SMILES: `C#Cc1ccc2sc(C(=O)NCCCCN3CCN(c4ccccc4OC)CC3)cc2c1`

Prediction: 98.6% confidence (Class 1 = Active)

Trustworthiness Score: 67.3%

Top 5 Critical Bonds (Positive Contribution to Activity):

Rank	Bond Type	Environment	Shapley Value	Chemical Role
1	C-C (piperazine ring)	N-CH_2-CH_2-N	+0.242	⭐ **Key scaffold** - The saturated nitrogen-containing ring is the strongest positive contributor
2	C-C (piperazine ring)	N-CH_2-CH_2-N	+0.235	⭐ **Key scaffold** - Complementary bonds in the same ring system
3	C-N (amine linkage)	C-N-C	+0.142	⭐ **Linker** - Connects the piperazine to the carbonyl chain
4	N-C (aniline bond)	c-N-C	+0.118	⭐ **Aromatic attachment** - Connects piperazine to phenyl ring
5	N-C (amide chain)	N-C-C	+0.092	⭐ **Flexible linker** - The butyl chain between amide and piperazine

---

2. INACTIVE Compound (Class 0)

SMILES: `COc1cccc(-c2ccc3c(NC(=O)c4ccc(C(=O)NO)cc4)n[nH]c3c2)c1F`

Prediction: 99.6% confidence (Class 0 = Inactive)

Trustworthiness Score: 92.4%

Top 5 Critical Bonds (Positive Contribution to Inactivity):

Rank	Bond Type	Environment	Shapley Value	Chemical Role
1	C=O (amide carbonyl)	c-C(=O)N	+0.056	🔬 **Hydrogen bond acceptor** - Strong carbonyl group
2	C-C (aryl-aryl)	c-C-c	+0.048	🔬 **Extended conjugation** - Links aromatic systems
3	C-O (methoxy)	C-O-c	+0.046	🔬 **Electron donor** - Methoxy substituent on phenyl
4	C-C (aryl-aryl)	c-C-c	+0.041	🔬 **Scaffold backbone** - Core aromatic linkage
5	C=O (nitrocarbonyl)	C(=O)NO	+0.036	🔬 **Electron-withdrawing** - Nitro carbonyl group

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📊 Key Differences: Active vs Inactive

Structural Features Driving Activity:

Feature	Active Compound	Inactive Compound	Impact
Core Scaffold	**Saturated piperazine ring** (N-containing)	Extended fused aromatic system	✅ **Positive for activity**
Linker Type	**Flexible aliphatic chain** (butyl)	Rigid aromatic linkages	✅ **Positive for activity**
Nitrogen Content	**Multiple amines** (3+ N atoms)	Single amide nitrogen	✅ **Positive for activity**
Aromaticity	**Moderate** (2 aromatic rings)	**High** (4+ fused/aromatic rings)	❌ **Negative for activity**
Molecular Flexibility	**High** (rotatable bonds)	**Low** (rigid planar structure)	✅ **Positive for activity**
Substituents	**Methoxy** (electron-donating)	**Nitro** (electron-withdrawing)	⚠️ **Context-dependent**

🧬 Chemical Interpretation

Why the Active Compound Works:

1. 🎯 **Piperazine Scaffold**: The saturated N-heterocyclic ring (piperazine) provides:
   - **Basic nitrogen atoms** for ionic interactions with biological targets
   - **Three-dimensional structure** enabling better binding pocket accommodation
   - **Optimal spacing** between pharmacophores
2. 🔗 **Flexible Linker**: The butyl chain (CH_2-CH_2-CH_2-CH_2) between the amide and piperazine:
   - Provides **conformational freedom** to adopt optimal binding pose
   - Allows **distance tuning** between pharmacophoric groups
   - Increases **lipophilicity** for membrane permeability
3. 💊 **Balanced Aromaticity**: Only 2 aromatic rings with a methoxy group:
   - Maintains **optimal logP** for drug-likeness
   - Provides **π-stacking** capability without excessive rigidity

Why the Inactive Compound Fails:

1. 🔒 **Too Rigid**: The fused aromatic system (indole-like + phenyl + phenyl) creates:
   - **Planar, inflexible structure** that cannot adapt to binding pockets
   - **High aromaticity** leading to poor solubility and non-specific binding
2. ⚡ **Electron-Withdrawing Groups**: The nitro carbonyl (C(=O)NO) and extended conjugation:
   - **Reduces basicity** of nearby nitrogen atoms
   - Creates **electronic imbalance** for target interactions
3. 📏 **Extended Conjugation**: Multiple aromatic rings linked together:
   - Increases **molecular weight** beyond optimal range

- Creates **hydrophobic patches** that can cause aggregation

📈 Model Performance Insights

Metric	Active Compound	Inactive Compound
Prediction Confidence	98.6%	99.6%
Trustworthiness	67.3%	92.4%
Key Structural Signal	Piperazine ring + flexible linker	Extended aromatic system
Edge Importance Pattern	Concentrated on N-heterocycle	Spread across aromatic bonds

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🎯 Takeaway Messages

✅ **The GCN model correctly learned that:**

- **Saturated N-heterocycles** (like piperazine) are strong positive indicators of activity
- **Flexible aliphatic linkers** enhance binding potential
- **Extended aromatic systems** with electron-withdrawing groups correlate with inactivity

✅ **EdgeSHAPer revealed:**

- **Active compounds** have concentrated importance on the piperazine scaffold
- **Inactive compounds** have spread importance across aromatic bonds
- The model distinguishes activity based on **3D vs 2D structural features**



